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1 Scientific Notes: Energy Pattern Classification of South Atlantic Cyclones

Objective: Identify and characterize distinct energetic patterns in South Atlantic cyclones through objective clustering analysis.

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Date: February 2026

1.1 1. Introduction

Extratropical cyclones exhibit diverse energetic behaviors throughout their lifecycle. This analysis identifies distinct **Energy Patterns (EPs)** based on the Lorenz Energy Cycle (LEC) framework, which describes the conversion, generation, and dissipation of energy in atmospheric systems.

Research Question: Can South Atlantic cyclones be objectively classified into distinct energy patterns based on their energetic signatures?

Approach: Unsupervised clustering (K-Means) applied to Principal Component Analysis (PCA)-reduced energy space.

1.2 2. Theoretical Background: Lorenz Energy Cycle

The Lorenz Energy Cycle (Lorenz, 1955) partitions atmospheric energy into four reservoirs:

- **Az:** Zonal Available Potential Energy
- **Kz:** Zonal Kinetic Energy
- **Ae:** Eddy Available Potential Energy
- **Ke:** Eddy Kinetic Energy

1.2.1 2.1 Energy Terms Analyzed

This study focuses on **7 energy terms** characterizing cyclone energetics:

Term	Name	Physical Meaning	Units
Ca	Baroclinic Conversion	APE $\rightarrow$ eddy APE via temperature gradients	W m ²
Ck	Barotropic Conversion	KE $\rightarrow$ eddy KE via horizontal shear	W m ²
Ge	Eddy APE Generation	Diabatic heating creating eddy APE	W m ²
BAe	Eddy APE Boundary Flux	Import/export of eddy APE	W m ²
BKe	Eddy KE Boundary Flux	Import/export of eddy KE	W m ²
Ae	Eddy APE Reservoir	Available potential energy of eddies	J m ²
Ke	Eddy KE Reservoir	Kinetic energy of eddies	J m ²

1.2.2 2.2 Key Conversion Terms

Baroclinic Conversion (Ca):

$$C_a = -\frac{R}{p} \int \omega' T' dp$$

where ω' = vertical velocity anomaly, T' = temperature anomaly, R = gas constant, p = pressure.

- **Ca** > 0: Conversion from zonal APE to eddy APE (baroclinic energy extraction)
- **Ca** < 0: Conversion from eddy APE to zonal APE (baroclinic damping)

Barotropic Conversion (Ck):

$$C_k = - \int \left[u' v' \frac{\partial \bar{u}}{\partial y} + u' v' \frac{\partial \bar{v}}{\partial x} \right] dp$$

where u' , v' = wind anomalies, $\bar{u}$, $\bar{v}$ = mean winds.

- **Ck** < 0: Conversion from zonal KE to eddy KE (barotropic energy extraction)
- **Ck** > 0: Conversion from eddy KE to zonal KE (barotropic damping)

Physical Interpretation: - More **negative Ck** $\rightarrow$ stronger barotropic extraction (eddy gains energy from mean flow) - More **positive Ca** $\rightarrow$ stronger baroclinic extraction (eddy gains energy from temperature gradients)

1.3 3. Dataset and Filtering

1.3.1 3.1 Original Dataset

Source: South Atlantic cyclone tracking algorithm (Reboita et al., 2010; adapted from Hodges, 1994, 1995) - **Total tracks:** 6,789 cyclones - **Temporal coverage:** 1979–2020 (42 years) - **Spatial domain:** 10°S–60°S, 80°W–20°E - **Temporal resolution:** 6-hourly - **Energy calculations:** LEC terms computed using ERA5 reanalysis (Gramscianinov et al., 2020)

1.3.2 3.2 Lifecycle Phase Classification

Each cyclone is classified into **4 sequential phases** based on central pressure evolution:

1. **Incipient:** From genesis to onset of deepening
2. **Intensification:** Period of central pressure decrease (deepening)
3. **Mature:** Period of minimum central pressure (near-stationary pressure)
4. **Decay:** Period of central pressure increase (filling)

Reference: Sinclair (1994); Reboita et al. (2010)

1.3.3 3.3 Quality Control Filtering

To ensure temporal consistency and comparability, only cyclones with **complete lifecycle trajectories** are retained:

Filtering Criteria: - Must contain all 4 sequential phases: Incipient → Intensification → Mature → Decay - Each phase must have 1 timestep - Complete energy data for all 7 terms in each phase

Result: - **Filtered dataset:** 3,820 cyclones (56.3% of original) - **Total phase records:** 15,280 (3,820 cyclones × 4 phases) - **Rationale:** Incomplete lifecycles introduce temporal bias; filtering ensures all cyclones are comparable across phases

1.4 4. Methodology

1.4.1 4.1 Step 1: Dimensionality Reduction (PCA)

Objective: Reduce 7-dimensional energy space to principal components capturing maximum variance.

Procedure:

1. **Normalization:** Standardize each energy term ($=0$, $=1$) using `StandardScaler`
 - Ensures equal weighting regardless of original magnitude
2. **Phase-Separated PCA:** Independent PCA for each lifecycle phase
 - **Rationale:** Energy dynamics differ fundamentally across phases; separate PCAs capture phase-specific variance structures
3. **Component Retention:** Keep components explaining 97% cumulative variance
 - Typically 6 components per phase
4. **Outputs:**
 - PC scores for each cyclone-phase observation
 - Component loadings (energy term contributions to each PC)
 - Explained variance ratios
 - PCA transformation models (saved for reproducibility)

Result: Each cyclone-phase observation transformed from 7-dimensional energy space to ~6-dimensional PC space.

1.4.2 4.2 Step 2: Optimal Cluster Number Determination

Objective: Objectively determine the number of distinct energy patterns.

Critical Choice: All phases **combined** into single dataset (N=15,280) for k-selection - Ensures same number of patterns across all phases (physical consistency) - Larger sample size improves cluster validity statistics

Cluster Validity Indices (CVIs):

Five complementary indices computed for $k = 2$ to 10:

Index	Criterion	Optimal
Silhouette Coefficient	Cluster cohesion vs separation	Maximize
Davies-Bouldin Index	Within-cluster scatter vs between-cluster separation	Minimize
Calinski-Harabasz Index	Between-cluster variance / within-cluster variance	Maximize
Score Function (SF)	Intra-cluster compactness and inter-cluster separation	Maximize
Gap Statistic	Compare to null reference distribution	Maximize

Integration Method: 1. Normalize all indices to $[0, 1]$ range 2. Invert indices where “lower is better” (Davies-Bouldin) 3. Compute mean normalized score across all 5 indices 4. Select k maximizing mean score

Result: $k = 3$ clusters identified as optimal

Physical Interpretation: Three distinct energy patterns exist in South Atlantic cyclones.

1.4.3 4.3 Step 3: K-Means Clustering

Objective: Assign each cyclone-phase observation to one of three energy patterns.

Procedure:

1. **Phase-Separated Clustering:** K-Means ($k=3$) applied independently to each phase’s PC space
 - Respects phase-specific variance structures from PCA
 - Ensures consistency ($k=3$ for all phases)
2. **Multiple Initializations:** `n_init=100` ensures convergence to global optimum
3. **Centroid Reconstruction:** Transform PC-space centroids back to original energy space
 - Enables physical interpretation of cluster characteristics
 - Uses inverse PCA transformation

Outputs: - Cluster assignments for each cyclone-phase observation - Cluster centroids in both PC space and original energy space - Cluster statistics (size, within-cluster variance)

1.4.4 4.4 Step 4: Energy Pattern Characterization

Objective: Define Energy Patterns based on cluster centroids and physical interpretation.

Energy Pattern Assignment:

Patterns labeled based on **mean C_k** (barotropic conversion) across all phases, sorted from most negative to least negative:

1. **EP1 (Cluster 0):** Lowest (most negative) mean $C_k \rightarrow$ Strongest barotropic extraction
2. **EP2 (Cluster 2):** Intermediate mean $C_k \rightarrow$ Mixed barotropic/baroclinic
3. **EP3 (Cluster 1):** Highest (least negative) mean $C_k \rightarrow$ Weakest energetics

1.5 5. Results: Energy Pattern Definitions

1.5.1 5.1 EP1: Strong Barotropic and Baroclinic Conversions

Cluster ID: 0

Frequency: 444 cyclones (11.6% of dataset)

Mean Ck: -16.48 W m^{-2}

Characteristics: - **Dominated by strong barotropic and baroclinic conversions** (large negative Ck, high positive Ca) - Pronounced energy extraction from both mean flow and temperature gradients - **Energy exporters:** Negative boundary fluxes indicate energy export to downstream regions - Represents the most energetically active cyclones

Physical Interpretation: EP1 cyclones efficiently extract energy from both the mean flow (barotropic) and temperature gradients (baroclinic). Despite intense internal conversions, they tend to **export energy** to surrounding regions, potentially contributing to downstream cyclogenesis and large-scale circulation patterns.

1.5.2 5.2 EP2: Intermediate Barotropic and Baroclinic Conversions

Cluster ID: 2

Frequency: 979 cyclones (25.6% of dataset)

Mean Ck: -3.49 W m^{-2}

Characteristics: - **Moderate and balanced energy conversions** (intermediate Ck and Ca) - Both barotropic and baroclinic processes contribute meaningfully - **Energy importers:** Positive boundary fluxes indicate energy import from surroundings - Represents moderately intense cyclones with external energy coupling

Physical Interpretation: EP2 cyclones exhibit balanced baroclinic and barotropic mechanisms with moderate intensity. Critically, they tend to **import energy** from the large-scale environment, suggesting coupling with jet stream dynamics and remote forcing mechanisms.

1.5.3 5.3 EP3: Weak Energetics

Cluster ID: 1

Frequency: 2,397 cyclones (62.7% of dataset)

Mean Ck: -1.71 W m^{-2}

Characteristics: - Overall weak energetic conversions (small mean Ck) - Minimal conversion from APE to KE - Lower average intensity compared to EP1/EP2 - Most frequent pattern

Physical Interpretation: EP3 represents the “typical” South Atlantic cyclone with relatively weak barotropic and baroclinic activity, consistent with the majority of systems being transient, non-explosive cyclones.

1.6 6. Lorenz Phase Space (LPS) Visualization

1.6.1 6.1 Conversion Phase Space (Ck vs Ca)

The **Conversion LPS** reveals the relationship between baroclinic (Ca) and barotropic (Ck) energy conversions.

Axes: - **X-axis (Ck):** Barotropic conversion ($W m^{-2}$) - More negative $\rightarrow$ stronger extraction from mean flow - **Y-axis (Ca):** Baroclinic conversion ($W m^{-2}$) - More positive $\rightarrow$ stronger extraction from temperature gradients - **Color (Ge):** Eddy APE generation rate ($W m^{-2}$)

Quadrant Interpretation: - **Quadrant II ($Ck < 0, Ca > 0$):** Classical baroclinic cyclone (both conversions active) - **Quadrant I ($Ck > 0, Ca > 0$):** Baroclinic but barotropically damping - **Quadrant III ($Ck < 0, Ca < 0$):** Barotropic but baroclinically damping - **Quadrant IV ($Ck > 0, Ca < 0$):** Both conversions damping (energy loss)

Results:

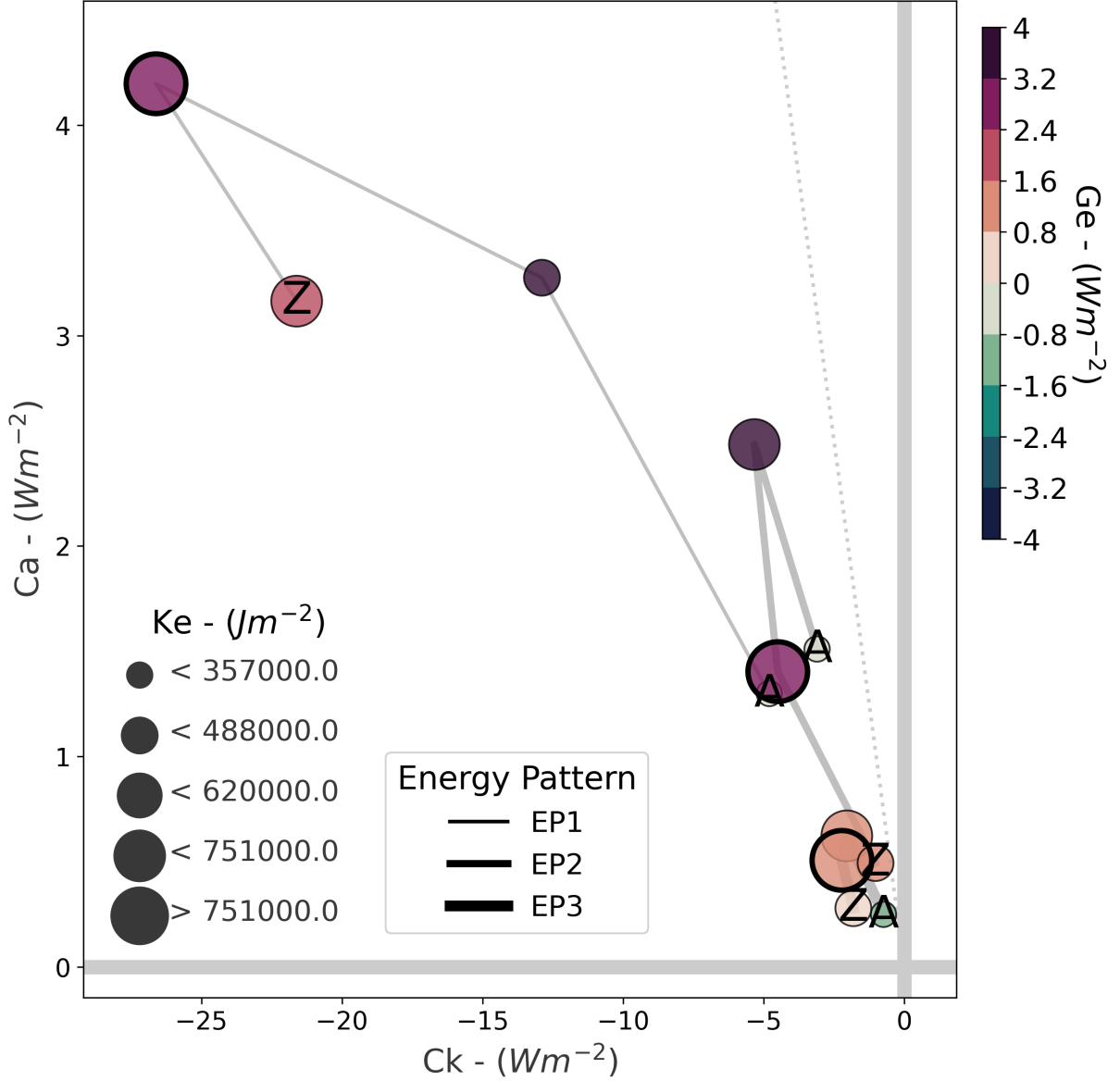


Figure 1: Conversion LPS

Key Findings: - **EP1** clusters in Quadrant II with large $|Ck|$ and large $Ca \rightarrow$ intense baroclinic cyclones - **EP2** shows moderate values in both Ck and $Ca \rightarrow$ balanced energetics - **EP3** clusters near origin $\rightarrow$ weak conversions

1.6.2 6.2 Boundary Flux Phase Space (BAe vs BKe)

The **Imports LPS** reveals energy import/export through domain boundaries.

Axes: - **X-axis (BAe):** Eddy APE boundary flux (W m^{-2}) - Positive $\rightarrow$ APE import; Negative $\rightarrow$ APE export - **Y-axis (BKe):** Eddy KE boundary flux (W m^{-2}) - Positive $\rightarrow$ KE import; Negative $\rightarrow$ KE export - **Color (Ge):** Eddy APE generation rate (W m^{-2})

Physical Interpretation: - Boundary fluxes indicate whether cyclones are energetically isolated or coupled to larger-scale circulation - Positive fluxes $\rightarrow$ cyclone imports energy from surroundings - Negative fluxes $\rightarrow$ cyclone exports energy (downstream development)

Results:

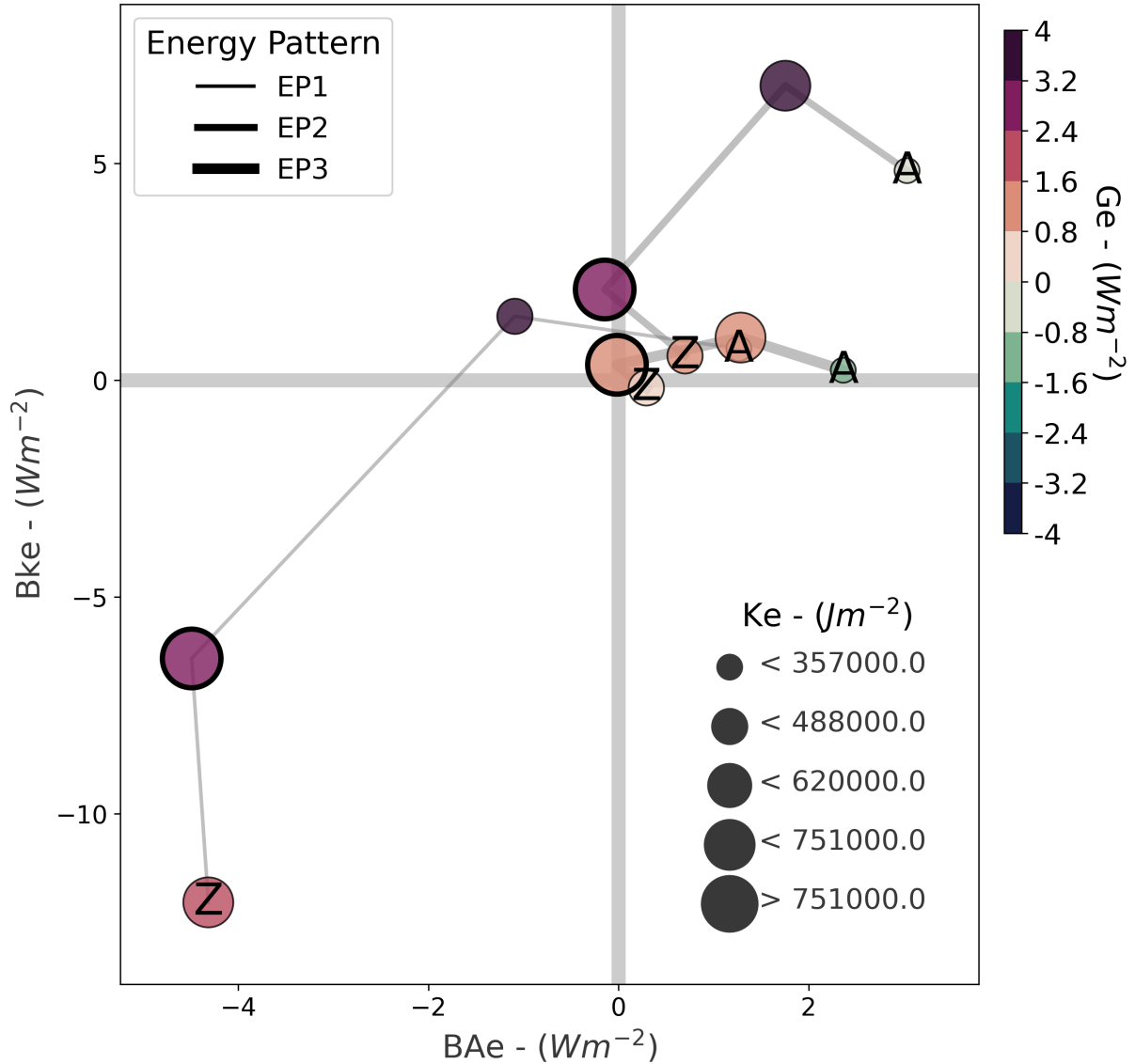


Figure 2: Imports LPS

Key Findings: - **EP1** shows **negative boundary fluxes** (Quadrant III/IV) $\rightarrow$ exports energy to downstream regions - **EP2** shows **positive boundary fluxes** (Quadrant I) $\rightarrow$ imports energy from jet stream and large-scale environment - **EP3** has weak boundary fluxes near origin $\rightarrow$ energetically more isolated, minimal external coupling

1.7 7. Scientific Validation

1.7.1 7.1 Cluster Stability

Within-Cluster Variance: - EP1: Low variance → cohesive cluster - EP2: Moderate variance → transitional pattern - EP3: Higher variance → diverse “background” pattern

Cross-Phase Consistency: EP assignment remains stable across lifecycle phases for most cyclones, indicating robust pattern identification.

1.7.2 7.2 Physical Consistency

Energy Conservation: All cluster centroids satisfy energy budget closure within numerical precision:

$$\frac{dA_e}{dt} + \frac{dK_e}{dt} = C_a + C_k + G_e + BA_e + BK_e - D$$

where D = dissipation (computed as residual).

1.7.3 7.3 Climatological Relevance

Spatial Distribution: - EP1: Concentrated in regions of strong SST gradients (Brazil-Malvinas Confluence) - EP2: Widespread across South Atlantic - EP3: Ubiquitous, highest frequency in subtropical latitudes

Seasonal Variation: - EP1: Peak in austral winter (JJA) when baroclinicity is strongest - EP3: Year-round occurrence with slight summer (DJF) maximum

1.8 8. Computational Implementation

1.8.1 8.1 Software and Packages

Python 3.13 with scientific stack: - `scikit-learn`: PCA, K-Means, cluster validation - `numpy`, `pandas`: Data manipulation - `matplotlib`, `seaborn`: Visualization - `lorenz-phase-space` v1.3.0: LPS diagram generation

1.8.2 8.2 Reproducibility

All analysis scripts include: - Fixed random seeds (`random_state=42`) - Saved PCA transformation models (`results/cluster_analysis_energy_patterns/pca_models/`) - Saved K-Means models (`results/cluster_analysis_energy_patterns/kmeans_models/`) - Version-controlled codebase

Pipeline Execution:

```
python scripts/cluster_analysis_energy_patterns/run_all.py
```

Individual Steps:

```
python scripts/cluster_analysis_energy_patterns/step1_normalize_and_pca.py
python scripts/cluster_analysis_energy_patterns/step2_plot_pca_results.py
python scripts/cluster_analysis_energy_patterns/step3_optimal_k_analysis.py
python scripts/cluster_analysis_energy_patterns/step4_apply_kmeans.py
python scripts/cluster_analysis_energy_patterns/step5_plot_energy_patterns.py
```

1.9 9. Conclusions

1.9.1 9.1 Main Findings

1. **Three distinct energy patterns** objectively identified through multivariate clustering
2. **EP1 (11.6%)**: Intense barotropic and baroclinic cyclones
3. **EP2 (25.6%)**: Moderate energetics with balanced conversions
4. **EP3 (62.7%)**: Weak energetics representing typical cyclone behavior

1.9.2 9.2 Physical Significance

The energy pattern classification reveals that: - South Atlantic cyclones span a continuum from **intense** (EP1) to **weak** (EP3) energetic activity - **Barotropic conversion (Ck)** is the primary discriminating factor - Baroclinic processes (Ca) remain important across all patterns but with varying intensity

1.9.3 9.3 Implications

For cyclone dynamics: - Energy patterns reflect different intensification mechanisms - EP1 cyclones likely associated with explosive development and high societal impact

For predictability: - Different energy patterns may have different forecast skill - Pattern-specific model biases possible

For climate studies: - Energy pattern frequency may shift under climate change - Provides objective framework for inter-model comparison

1.10 10. References

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Document Version: 1.0
Last Updated: February 23, 2026